Product Requirements Document (PRD)

1. Product name

SwarRakshak

For this PRD, the product is called **SwarRakshak** because the name is short, memorable, and communicates protection against voice-based attacks.

2. One-line product description

SwarRakshak is a privacy-aware voice-risk assistant that analyzes call audio and request context to warn users about possible voice-cloning impersonation before they approve a sensitive action.

3. Problem statement

Artificial intelligence can now generate a voice that sounds like a real executive, customer, government officer, or family member. An attacker can use this cloned voice, together with leaked personal information, to make a convincing phone or online call.

During the call, the attacker may request a fund transfer, password, confidential document, account change, or urgent approval. Caller ID is not reliable proof of identity, and human familiarity with a voice is no longer sufficient. Employees under time pressure may also skip normal verification procedures.

Most existing defenses are separate from the call. A company may have a callback policy, multifactor authentication, fraud rules, or voice biometrics, but the user may not receive a clear warning at the moment of risk. A detector that only says “real” or “fake” is also not enough because voice detection is probabilistic and a genuine human can still make a fraudulent request.

SwarRakshak addresses this gap by combining three types of evidence:

1. **Voice authenticity:** Does the audio contain patterns commonly associated with synthetic or manipulated speech?
2. **Speaker consistency:** Does the current voice resemble a trusted reference sample, when one is available?

3. **Call context:** Is the request urgent, unusual, coming from an unknown number, or connected to a high-value action?

The product provides an understandable risk level and recommends a safer next step. It supports decision-making; it does not replace independent identity verification.

4. Target users

Primary target user

The primary target user for the hackathon MVP is a **bank or company finance employee who receives voice instructions related to money transfers or sensitive account actions**.

This user is a good starting point because the consequences of a successful impersonation attack can be serious and the verification workflow is easy to demonstrate.

Secondary users

- Fraud and security analysts who investigate suspicious calls.
- Call-center agents who verify customers or change account details.
- Company executives and administrative staff who receive urgent requests.
- Government employees handling confidential instructions.
- Telecom and enterprise communication providers that may later integrate the service.

Initial user persona

Name: Priya, Finance Operations Executive

Situation: Priya receives a call from someone claiming to be the Chief Financial Officer. The caller knows the CFO’s name and asks her to approve an urgent transfer to a new account.

Current difficulty: The caller sounds convincing. Priya has to decide quickly, but the existing system does not provide an audio-risk warning.

Desired outcome: Priya receives a clear warning, understands why the call is risky, and knows how to verify the request without relying on the caller’s phone number.

5. User pain points

- A cloned voice can sound similar to someone the user knows.
- Attackers create urgency so that users do not stop to verify the request.

- Caller ID may be spoofed, hidden, or unavailable.
- Manual verification is useful but is often forgotten under pressure.
- Users cannot personally identify technical audio artifacts.
- A warning without an explanation may be ignored.
- False alarms can slow down legitimate business operations.
- Voice recordings may contain private conversations and confidential information.
- Background noise, poor microphones, network compression, illness, and accents can affect detection accuracy.
- A system that detects only synthetic speech may miss a fraudulent request made by a real human.

6. Proposed solution

SwarRakshak will be a web-based prototype that analyzes a short voice sample and structured call information. It will present the result as an explainable risk assessment.

How the prototype works

1. The user uploads a short current-call sample or records one through the browser.
2. The user optionally supplies a trusted reference sample for the person the caller claims to be.
3. The application preprocesses the audio and divides it into short segments.
4. An audio-analysis model estimates whether the speech may be synthetic or manipulated.
5. A speaker-comparison model estimates whether the current speaker resembles the reference sample.
6. The user enters simple context details, such as whether the number is known, whether the request is urgent, and whether it involves a large transfer.
7. A risk engine combines the signals into a score from 0 to 100.
8. The interface displays Low, Medium, or High risk, the main reasons, the confidence level, and a recommended action.
9. The application creates an incident summary without retaining the raw recording by default.

Important product principle

SwarRakshak must not claim that it can prove a caller is fake. Its wording should be cautious and useful:

> “This call has a high risk of impersonation. Pause the requested action and verify the caller using an independently trusted channel.”

7. Product goals

Primary goals

1. **Detect risk early:** Analyze short audio segments quickly enough to provide a warning during a simulated live call.
2. **Combine multiple signals:** Use audio, speaker consistency, and call context instead of depending on one detector.
3. **Give an actionable recommendation:** Tell the user what to do next when the risk is Medium or High.
4. **Explain the result:** Show the main factors that contributed to the risk score.
5. **Protect privacy:** Delete raw audio after analysis by default and retain only the minimum necessary metadata.
6. **Deliver a convincing hackathon demo:** Demonstrate different outcomes for genuine, synthetic, and contextually suspicious scenarios.
7. **Keep the implementation realistic:** Use uploaded or browser-recorded audio rather than attempting direct integration with telecom networks.

Non-goals for the MVP

The MVP will not attempt to provide perfect deepfake detection, automatic transaction blocking, universal language support, or production-grade telecom integration.

8. User stories

Core user stories

1. **As a finance employee,** I want to upload or record a call sample so that SwarRakshak can analyze it.
2. **As a finance employee,** I want to provide a trusted reference voice when available so that the system can compare the current speaker with the claimed identity.
3. **As a finance employee,** I want to enter basic call context so that the risk assessment includes the danger of the requested action.

4. **As a finance employee,** I want to see a Low, Medium, or High risk level so that I can understand the result quickly.
5. **As a finance employee,** I want to see why a call was flagged so that I can judge whether the warning is reasonable.
6. **As a finance employee,** I want a recommended verification step so that I know how to respond safely.
7. **As a finance employee,** I want the system to analyze audio in short segments so that the risk can update during a call.
8. **As a privacy-conscious organization,** I want raw audio deleted after analysis so that sensitive conversations are not stored unnecessarily.
9. **As a fraud analyst,** I want to view a summary of flagged events so that I can investigate without immediately listening to every recording.
10. **As a product administrator,** I want risk thresholds to be configurable so that different workflows can use different levels of caution.

User stories intentionally deferred

- As a telecom operator, I want to inspect every call across the network.
- As a bank, I want SwarRakshak to automatically reject a payment.
- As a global organization, I want support for every language and dialect.

These are possible future goals, not hackathon MVP requirements.

9. Functional requirements

FR-1: Audio input

The system must allow the user to upload a short audio file in a common format such as WAV, MP3, or M4A. If time permits, it may also support browser microphone recording.

FR-2: Reference sample input

The system should optionally accept a consented reference sample for the person the caller claims to be. The reference sample must not be required because some scenarios will not have one.

FR-3: Audio preprocessing

The system must convert supported audio files to a standard internal format, normalize basic properties, and split the audio into short analysis segments.

FR-4: Voice authenticity analysis

The system must return an estimate of whether the audio contains patterns associated with synthetic or manipulated speech. The result must be represented as evidence or probability, not as an absolute fact.

FR-5: Speaker comparison

When a reference sample is provided, the system should calculate a similarity result between the current sample and the reference sample. If no reference is available, this signal must be marked as unavailable rather than treated as a negative result.

FR-6: Call-context form

The system must allow the user to enter or select simple context fields:

- Known or unknown caller number.
- Normal or urgent request.
- Low, medium, or high transaction value.
- Normal procedure or request to bypass procedure.
- Known contact, vendor, executive, customer, or unknown relationship.

FR-7: Risk scoring

The system must combine available signals into a score from 0 to 100. The initial prototype may use a transparent weighted-rule engine.

Suggested prototype weights:

Signal	Example contribution
Synthetic-voice indication	Up to 40 points
Speaker mismatch	Up to 30 points
Unknown caller number	10 points
High-value request	10 points
Urgency or policy bypass	Up to 10 points

The weights are starting values for the demo and must not be presented as scientifically validated production thresholds.

FR-8: Risk-level classification

The system must map the score to a simple label. Initial thresholds may be:

- **Low risk:** 0–29
- **Medium risk:** 30–59
- **High risk:** 60–100

The thresholds must be easy to change in configuration.

FR-9: Explanation of result

The system must display at least two reason codes when evidence is available. Examples include:

- Audio contains patterns associated with synthetic speech.
- Current voice differs from the reference sample.
- Caller number is not in the trusted directory.
- Request is urgent and involves a high-value transfer.
- Caller is asking the employee to bypass normal approval steps.

FR-10: Recommended action

The system must recommend an action based on the risk level:

- **Low:** Continue normal verification procedures.

- **Medium:** Avoid sharing sensitive information and ask a verification question.
- **High:** Pause the action and call the person through an independently trusted number, or require second-person approval and multifactor authentication.

FR-11: Segment timeline

The system should show which audio segments were flagged, using a simple timeline or list. This feature demonstrates the near-real-time concept without requiring a live telecom integration.

FR-12: Incident summary

The system should allow the user to export or view a summary containing the event ID, timestamp, risk score, risk level, confidence, reason codes, and recommended action. The summary should not include raw audio by default.

FR-13: Privacy behavior

The system must delete temporary raw audio after analysis or provide a clear user-controlled retention option. The interface must state whether audio is being retained.

FR-14: Demo scenarios

The product must support at least three prepared scenarios:

1. Genuine voice with a normal request.
2. Synthetic or cloned voice with suspicious audio evidence.
3. Voice that appears normal but involves an unknown number and an urgent high-value request.

10. Non-functional requirements

NFR-1: Usability

A beginner user should be able to complete an analysis without technical knowledge. The main result should be understandable within a few seconds of completion.

NFR-2: Performance

For the hackathon demo, a short audio sample should produce a result within approximately 10 seconds on the chosen demo environment. For simulated streaming, the interface should update after each short segment where practical.

NFR-3: Explainability

Every Medium or High result should show human-readable reasons. The product must avoid showing only a model number with no explanation.

NFR-4: Privacy

Raw audio should not be retained by default. The system should collect only the fields needed for the demo and should use consented or synthetic audio.

NFR-5: Security

Uploaded files must be validated for type and size. The application must not execute uploaded files. Any retained metadata should be protected by normal application access controls.

NFR-6: Reliability

The application should handle invalid audio, missing reference samples, model errors, and unavailable analysis signals gracefully. It must show “Unable to determine” rather than inventing a confident result.

NFR-7: Accessibility

Use readable text, clear color labels, and text descriptions in addition to color. For example, High risk should not be represented only by a red color.

NFR-8: Scalability direction

The prototype should separate audio processing, model inference, and risk scoring conceptually so that the architecture could later support APIs and enterprise integrations. Full production scalability is not required for the hackathon.

NFR-9: Model transparency

The demo must identify whether it uses a pretrained model, a baseline classifier, or simulated sample results. The team must not claim validated production accuracy without evidence.

11. Core features

The core feature set for the hackathon is:

1. Audio upload.
2. Optional reference voice upload.

3. Audio preprocessing.
4. Synthetic-voice risk estimate.
5. Optional speaker similarity estimate.
6. Context form for call and transaction risk.
7. Transparent 0–100 risk score.
8. Low, Medium, or High label.
9. Explanation of contributing factors.
10. Recommended verification action.
11. Three prepared demo scenarios.
12. Temporary audio deletion and metadata-only incident summary.

If the team has limited time, features 1, 3, 4, 6, 7, 8, 9, and 10 are the minimum demonstration path. The reference comparison and timeline can be implemented next.

12. Nice-to-have features

These features are valuable but should not delay the core MVP:

- Browser microphone input with live segment updates.
- Speaker similarity against a stored, consented reference profile.
- Speech-to-text transcription.
- Detection of suspicious intent, such as requests for passwords or urgent payments.
- Support for Hindi and one additional Indian language.
- Accent-aware evaluation.
- Contact-directory integration.
- SMS, email, or in-app alerts.
- Configurable rules for different departments.
- Investigator dashboard with trends and filters.
- Feedback buttons for marking results as genuine or suspicious.
- On-device or edge inference.
- Detection of replayed recordings.

13. User journeys

Journey A: Genuine normal call

1. Priya opens SwarRakshak.
2. She uploads a consented reference sample and a current call sample.
3. She selects “Known number,” “Normal request,” and “Normal transaction.”
4. The audio analysis finds no strong synthetic-voice evidence.
5. The speaker similarity result is consistent with the reference.
6. The system displays **Low risk**.
7. The system tells Priya to continue normal company verification procedures.

Expected product value: The system gives reassurance without claiming certainty.

Journey B: Cloned voice requesting a transfer

1. Priya receives a call from someone claiming to be the CFO.
2. She uploads or records a short call sample.
3. She selects “Unknown number,” “Urgent request,” and “High-value transfer.”
4. The audio analysis identifies patterns associated with synthetic speech.
5. The current voice differs from the reference sample.
6. The system displays **High risk** with clear reasons.
7. The system recommends pausing the transfer and calling the CFO using a trusted number from the company directory.

Expected product value: The warning interrupts a potentially dangerous decision and gives Priya a specific next step.

Journey C: Real voice but suspicious request

1. Priya receives a call from a known person’s number.
2. The voice appears natural and the reference comparison is reasonably consistent.
3. The caller requests a large transfer urgently and asks Priya to bypass a second approval.
4. The context rules increase the risk score.
5. The system displays **Medium or High risk**, depending on the configured thresholds.

6. It recommends independent verification and second-person approval.

Expected product value: The product detects risky behavior even when the speaker is not obviously synthetic.

Journey D: Unable to determine

1. The user uploads a very short, noisy, or unsupported audio file.
2. The system cannot calculate reliable audio or speaker evidence.
3. The system explains that the result is inconclusive.
4. It recommends using an independent verification method instead of treating the call as safe.

Expected product value: The product avoids false certainty when the evidence is weak.

14. MVP scope

MVP objective

Demonstrate that a voice-cloning risk assistant can combine audio evidence and call context to recommend a safer action before a sensitive transaction is approved.

MVP included

- A simple web interface.
- Audio file upload.
- Optional reference sample upload.
- Audio preprocessing.
- A pretrained or prototype audio-analysis component.
- Optional speaker comparison.
- A structured context form.
- A transparent risk-scoring engine.
- Low, Medium, and High risk labels.
- Explanation text and reason codes.
- Recommended verification action.
- Three prepared demo scenarios.

- Basic privacy behavior, including temporary audio cleanup.
- A metadata-only incident summary.

MVP implementation approach

The team may simulate live operation by processing audio in short sequential chunks. It does not need to connect to a real phone carrier, bank, or enterprise collaboration platform.

If a reliable synthetic-speech model cannot be integrated during the hackathon, the team should use a clearly labeled baseline classifier or prepared demonstration data. The presentation must explain the prototype's limitations honestly.

Suggested MVP technology stack

- **Frontend:** React, or plain HTML/CSS/JavaScript for a simpler build.
- **Backend:** Python with FastAPI or Flask.
- **Audio processing:** FFmpeg and librosa.
- **Machine learning:** PyTorch, torchaudio, scikit-learn, or a suitable pretrained audio model.
- **Optional transcription:** Whisper or another speech-to-text model.
- **Storage:** SQLite for event metadata only.
- **Deployment:** Local machine, Docker, or a simple hackathon hosting environment.

MVP demo script

The demo should take no more than three to five minutes:

1. Introduce the problem with the fake executive call example.
2. Run the genuine normal-call scenario.
3. Run the cloned-voice high-risk scenario.
4. Run the natural-voice but suspicious-context scenario.
5. Show that the system explains its result and recommends an independent callback.
6. End by explaining privacy protections and model limitations.

15. Out-of-scope features

The following features are explicitly outside the MVP:

- Direct integration with mobile carriers or telecom signaling systems.

- Integration with every VoIP and collaboration platform.
- Automatic approval, rejection, or movement of real money.
- Automatic blocking or termination of calls.
- A production-grade detector trained on millions of real conversations.
- A claim of 100% detection accuracy.
- Complete support for every Indian language, dialect, and accent.
- Permanent storage of real people’s voice recordings without consent.
- A full enterprise identity, permissions, and compliance platform.
- Real banking-core integration.
- Automatic legal, employment, government, or financial decisions.
- A large analytics dashboard before the basic workflow works.
- Custom neural-network research unless the team already has the required expertise.

16. Success metrics

Because this is a hackathon MVP, success should be measured by usefulness, clarity, and demonstration quality rather than by unsupported accuracy claims.

Product and demo metrics

Metric	MVP target
Prepared scenarios that produce the intended risk category	At least 3 of 3
Scenarios with at least two understandable reasons	100% of flagged scenarios
High-risk scenario that produces a clear verification action	100% of the main demo runs
Time to analyze a short sample in the demo environment	Approximately 10 seconds or less
Raw audio retained after analysis by default	No
User can complete the main flow without developer assistance	Yes
Risk result visible without interpreting technical output	Yes

Technical evaluation metrics

If the team has enough labeled test samples, it may report:

- Accuracy, precision, recall, and false-positive rate on the prepared test set.
- Performance separately for genuine and synthetic samples.
- Performance under noise or compression.
- Speaker-comparison results with and without a reference sample.

The team must state the size and limitations of the test set. Results from a small hackathon dataset should not be presented as production performance.

Qualitative success criteria

Judges or test users should be able to answer “yes” to these questions:

- Is the problem immediately understandable?
- Does the product warn the user before a risky action?
- Does it explain why the call was flagged?

- Does it recommend a practical verification step?
- Does the team acknowledge uncertainty and privacy concerns?
- Is the MVP believable for the available development time?

17. Risks and assumptions

Product risks

Risk	Impact	Mitigation
The detector produces false positives	Users may distrust or ignore warnings	Show confidence and reasons; recommend verification rather than automatic blocking
The detector misses a new type of synthetic voice	A fraudulent call may appear safe	Combine audio evidence with context; avoid claiming perfect detection
Poor audio quality hides useful features	Analysis becomes unreliable	Show “inconclusive,” use short quality checks, and rely more on independent verification
Accents or languages affect results	Genuine users may be unfairly flagged	Test on diverse consented samples; avoid making unsupported language claims
Users treat the score as proof of identity	A dangerous decision may still be made	Use cautious wording and make verification recommendations prominent
Raw audio is exposed or retained unnecessarily	Privacy harm	Delete temporary files and store metadata rather than recordings
Model license is unsuitable	The team cannot legally use the model	Check the model and dataset license before integration
The integration takes too long	The demo is incomplete	Use uploaded audio and simulated live processing instead of telecom integration

Technical assumptions

- The demo has access to consented or synthetic audio samples.
- The chosen model can process short audio files in the available environment.
- The team can run a basic web application and backend API.
- Internet or package access is available during setup, or dependencies are prepared in advance.
- Context information can be entered manually for the MVP.

Data and privacy assumptions

- The team will not use real confidential bank calls or real financial credentials.
- Any human voice sample used in the demo will be recorded with permission.
- Test data will be deleted after the event unless retention is explicitly necessary and authorized.
- The product will show that a voice sample is sensitive personal data.

Ethical and safety assumptions

SwarRakshak is a decision-support tool, not a final identity authority. It must not be used by the prototype to make irreversible financial, legal, employment, medical, or government decisions automatically.

The team should avoid presenting a low score as proof that a caller is genuine. The safe interpretation is:

> **A high score means the user should slow down and verify independently. A low score means the available evidence did not raise a strong warning; it does not prove that the caller is trustworthy.**

Final product recommendation

For the hackathon, SwarRakshak should focus on one clear promise:

> **When a voice can no longer be trusted by itself, SwarRakshak combines voice authenticity, speaker consistency, and request context to recommend the safest next action before fraud occurs.**

A small, explainable, privacy-aware prototype will be stronger than an unfinished platform claiming to serve every telecom operator, language, and financial institution.